

Cross-Lingual Persona Stability in Small Language Models

An IPIP-50 Evaluation in English and Burmese

Prepared by

Ye Thu Aung

Two-Month Internship Final Research Report

August 2026

Independent study of bilingual psychometric response patterns

Abstract

Large language models can produce human-like self-descriptions, but an apparent personality profile may change when the same questionnaire is presented in another language. This study examined cross-lingual persona stability and response-format reliability in three compact open language models: Qwen 2.5 3B, Llama 3.2 3B, and Llama-SEA-LION v3.5 8B-R. A parallel English-Burmese version of the 50-item International Personality Item Pool (IPIP-50) questionnaire was administered in 20 independent sessions for every model-language condition. Each item requested one integer on a five-point Likert scale. The resulting design produced 6,000 item-level generations across 120 complete questionnaire sessions. Negatively keyed items were reverse-scored, and valid responses were aggregated into the Big Five domains of Extraversion, Agreeableness, Conscientiousness, Neuroticism, and Openness.

Response validity was the dominant cross-lingual result. All three models had a 0.0% parse-failure rate in English. In Burmese, Qwen failed on 921 of 1,000 responses (92.1%) and Llama failed on all 1,000 responses (100.0%). SEA-LION returned parseable responses for every item in both languages. For SEA-LION, the mean absolute difference between English and Burmese trait means was 0.14 points on the 1-5 scale; Agreeableness showed the largest difference at -0.43 points in Burmese. Because Qwen provided only 79 valid Burmese responses and Llama provided none, their Burmese trait profiles cannot support reliable persona comparisons. The study demonstrates that output compliance must be established before interpreting multilingual psychometric scores and suggests that regional language specialization may matter more than model size alone for this task.

Keywords: large language models; IPIP-50; Big Five; Burmese; multilingual evaluation; persona stability; instruction following

1. Introduction

1.1 Background

Large language models (LLMs) are increasingly used as conversational systems, assistants, and simulated agents. Their responses often appear to express stable preferences, attitudes, or personality characteristics. Recent research has therefore adapted human psychometric questionnaires to study personality-like patterns in model outputs [3, 4]. Such measurements can be useful for auditing interaction style and behavioral consistency, but they require careful interpretation: an LLM is generating text in response to a prompt, not reporting a human psychological state.

The Big Five framework organizes personality description into five broad domains: Extraversion, Agreeableness, Conscientiousness, Neuroticism, and Openness. Goldberg's work on Big Five factor markers provided a foundation for questionnaire-based measurement [1]. The International Personality Item Pool (IPIP) makes public-domain personality items available for research and provides a 50-item sample questionnaire with positive and negative scoring keys [2]. This instrument offers a compact, reproducible way to compare model response patterns.

Most LLM personality evaluations have concentrated on English or on comparatively large models. That leaves an important practical question for multilingual systems: does an apparent persona remain stable when the questionnaire language changes? Language can affect tokenization, instruction comprehension, cultural associations, and output formatting. For a lower-resource language such as Burmese, a model may fail to follow the questionnaire format before a meaningful trait score can even be calculated.

1.2 Problem statement

A cross-language difference in a computed trait mean can have at least two explanations. It may reflect a genuine change in the model's generated response pattern, or it may be an artifact caused by failure to understand the prompt and return a valid scale value. Aggregating sparse or malformed outputs can produce visually persuasive but methodologically weak profiles. Response reliability must therefore be analyzed alongside trait values.

The three models in this project provide a useful contrast. Qwen 2.5 3B and Llama 3.2 3B are compact general-purpose instruction models [5, 6]. Burmese is not among the eight officially supported languages listed for Llama 3.2 [6]. Llama-SEA-LION v3.5 8B-R was developed for Southeast Asian languages and its model card explicitly lists both Burmese and English among its supported languages [7]. The comparison is exploratory rather than a controlled test of model architecture because model families, sizes, training data, and regional specialization differ.

1.3 Research objectives

This internship project pursued three objectives:

- Build a reproducible, resumable pipeline for administering parallel English and Burmese IPIP-50 items to local open models.
- Measure output-format reliability for every model-language condition before interpreting personality scores.
- Compare English and Burmese Big Five profiles only where sufficient parseable responses exist, while documenting limitations for sparse conditions.

The study addressed two research questions. RQ1 asked whether the rate of valid one-to-five responses differed across model-language conditions. RQ2 asked whether the English and Burmese Big Five profiles were similar for models that produced complete data in both languages.

2. Methods

2.1 Research design

The study used a repeated, within-model evaluation design. Three models were each evaluated in two languages. Every model-language pair completed 20 sessions, and every session contained all 50 IPIP items. No conversational state was carried between items. The full grid was therefore:

Design element	Value
Models	3
Languages	2 (English and Burmese)
Sessions per model-language pair	20
Items per session	50
Complete questionnaire sessions	120
Total item-level generations	6,000

The experiment evaluated pretrained instruction models; no fine-tuning or parameter training was performed during this project.

2.2 Models and inference environment

Inference was performed locally through Ollama using the model identifiers recorded by the pipeline. The project compared two approximately 3-billion-parameter models with one regionally specialized 8-billion-parameter model.

Pipeline identifier	Short name	Research role
qwen2.5:3b	Qwen 2.5 3B	Compact general instruction model
llama3.2:3b	Llama 3.2 3B	Compact multilingual instruction model
aisingapore/Llama-SEA-LION-v3.5-8B-R	SEA-LION v3.5 8B-R	Southeast Asian language-specialized model

For every item, generation used a temperature of 0.7 and a maximum of eight predicted tokens. The

models were asked to output exactly one integer. The low output limit reduced unnecessary continuation while still allowing malformed outputs to be retained for auditing. Inference exceptions were logged and represented as missing values rather than terminating the full experiment.

2.3 Questionnaire and bilingual data

The source instrument was the 50-item IPIP sample questionnaire [2], derived from Big Five factor-marker research [1]. Each item was associated with one of five traits and a positive or negative scoring direction. The English and Burmese JSON files used matching item identifiers, trait labels, and keying directions. The Burmese file provided a parallel questionnaire in Burmese while retaining the same experimental structure.

The response scale ranged from 1 (Very Inaccurate) to 5 (Very Accurate). English prompts presented the English scale labels and statement. Burmese prompts presented the equivalent instructions, response scale, and translated statement. Both prompts explicitly required one number and prohibited punctuation, words, explanations, or introductory text.

2.4 Data collection procedure

The pipeline first validated and loaded 50 items for each language. It then iterated through model, language, session, and item. Every item was submitted through stateless generation so an answer to one statement could not influence later statements through conversation history. Raw text was stripped of surrounding whitespace and written immediately to a CSV file with session identifier, model name, language, item identifier, trait, keying direction, raw output, and parsed score.

The implementation was designed for a resource-constrained internship environment. Rows were flushed to disk after every generation. Before resuming, the pipeline counted records for every model-language-session key, retained only complete 50-item sessions without empty raw outputs, and reran interrupted sessions. This design protected long experiments from power, process, or model-service interruption while avoiding duplicate completed sessions.

2.5 Parsing and scoring

The parser searched each raw output for the first Western or Burmese numeral corresponding to 1 through 5. A response without one of these digits was marked as a parse failure. This approach is more permissive than exact-string matching because a digit embedded in a short phrase is accepted. Consequently, the reported failure rates indicate outputs with no recognizable valid scale digit; they do not measure perfect compliance with the instruction to return only one character.

Valid negatively keyed responses were reverse-scored using:

adjusted score = 6 - parsed score

Positively keyed responses retained their parsed value. Within each session, adjusted values were averaged for each trait. The analysis then calculated the mean, sample standard deviation, and number of contributing sessions for each model-language-trait combination. Parse failures were

separately counted over all 1,000 item attempts in each model-language condition.

2.6 Analysis strategy

Analysis emphasized descriptive statistics. First, item-level completeness and parse-failure rates established whether a condition could support interpretation. Second, trait means and between-session standard deviations described the response profile. Third, Burmese-minus-English deltas measured the direction and size of cross-language differences. Mean absolute language difference summarized the five deltas for a model only when both language profiles existed.

No inferential hypothesis test was applied. Conditions were not comparable after severe missingness, and the study was an exploratory internship project with one prompt design and one sampling configuration. Treating each generated item as an independent human observation would also overstate the sample size and violate the structure of the experiment.

3. Results

3.1 Dataset completeness

The final item-level file contained exactly 6,000 rows. All 120 model-language-session combinations contained 50 rows, confirming structural completeness of the generation grid. Structural completeness did not imply response validity: a completed row could contain raw text without a parseable score.

3.2 Parse-failure rates

All 3,000 English generations contained a recognizable score from 1 through 5. Burmese performance differed substantially by model. Qwen produced 79 valid Burmese scores and 921 failures. Llama produced no valid Burmese scores. SEA-LION produced 1,000 valid Burmese scores and no failures.

Model	English failures	Burmese failures	Burmese valid responses
Qwen 2.5 3B	0/1,000 (0.0%)	921/1,000 (92.1%)	79
Llama 3.2 3B	0/1,000 (0.0%)	1,000/1,000 (100.0%)	0
SEA-LION v3.5 8B-R	0/1,000 (0.0%)	0/1,000 (0.0%)	1,000

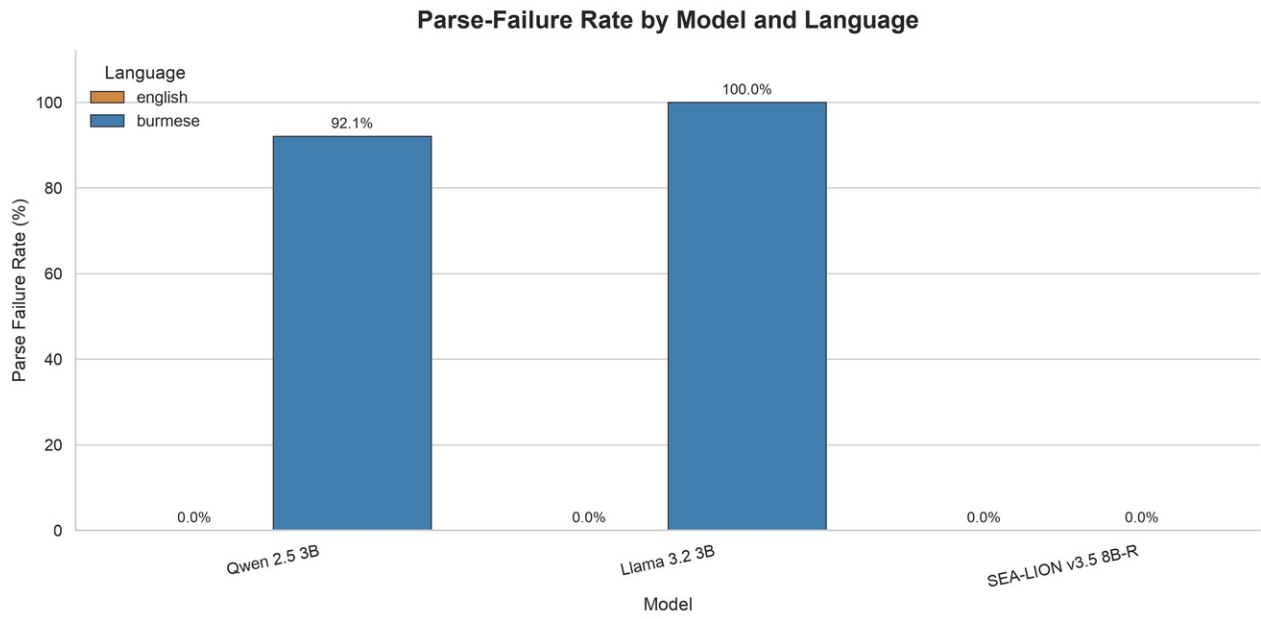


Figure 1. Parse-failure rate by model and questionnaire language.

3.3 Trait-score summaries

English trait profiles were available for all three models. SEA-LION also had complete Burmese data. Qwen's Burmese values were calculated from sparse valid items and had high between-session standard deviations; Llama had no Burmese trait scores. Figure 2 presents model-language means with error bars representing between-session standard deviation.

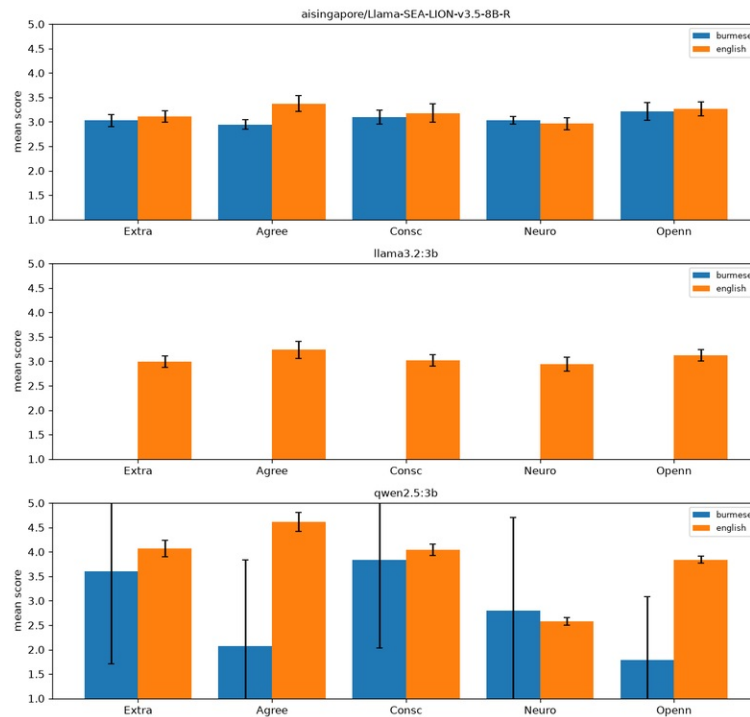


Figure 2. Mean reverse-scored Big Five values with between-session standard deviations.

3.4 SEA-LION cross-language profile

SEA-LION was the only model for which every English and Burmese response was parseable. Its English and Burmese means were close on four traits. Agreeableness had the largest absolute shift, decreasing from 3.375 in English to 2.945 in Burmese. The mean absolute difference across all five domains was 0.141 scale points.

Trait	English mean +/- SD	Burmese mean +/- SD	Burmese - English
Extraversion	3.105 +/- 0.115	3.030 +/- 0.122	-0.075
Agreeableness	3.375 +/- 0.162	2.945 +/- 0.095	-0.430
Conscientiousness	3.180 +/- 0.182	3.100 +/- 0.141	-0.080
Neuroticism	2.965 +/- 0.123	3.030 +/- 0.080	+0.065
Openness	3.265 +/- 0.146	3.210 +/- 0.180	-0.055

The radar plot in Figure 3 overlays the two language profiles. It also displays the sparse Qwen Burmese profile for transparency and explicitly marks the missing Llama Burmese condition.

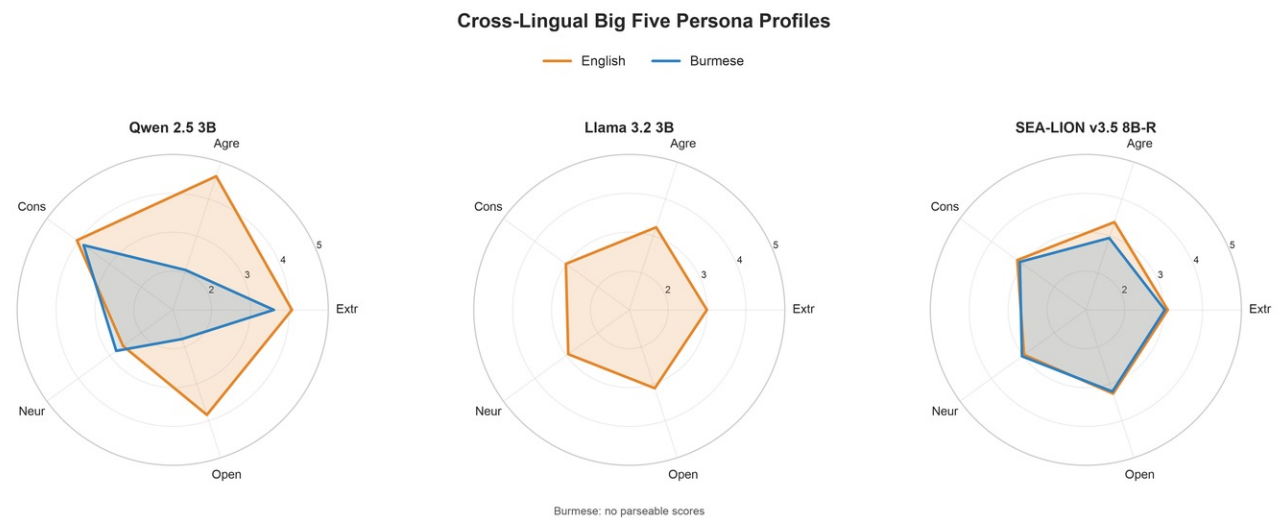


Figure 3. Cross-lingual Big Five profiles. *Qwen* Burmese values are sparse; *Llama* has no parseable Burmese scores.

Figure 4 expresses the same comparison as Burmese-minus-English deltas. Missing *Llama* values are shown as not available. *Qwen* deltas are visualized because they exist in the generated summary, but they should not be treated as reliable cross-language effects due to the 92.1% item-level failure rate.

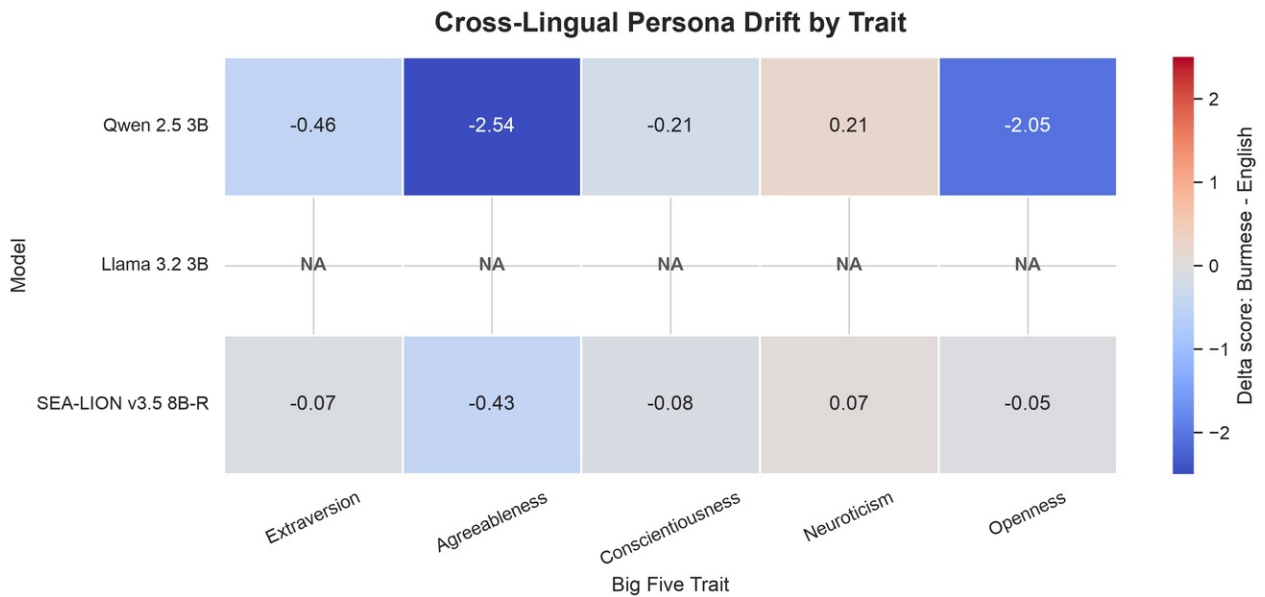


Figure 4. Burmese-minus-English trait deltas. NA indicates that no cross-language comparison could be computed.

4. Discussion

4.1 Interpretation of the findings

The strongest result was not a personality score; it was the separation between successful English response formatting and failed Burmese response formatting. All models handled the English questionnaire, while only SEA-LION completed the Burmese questionnaire without parse failures. This establishes a necessary ordering for multilingual persona evaluation: researchers should first test whether a model can understand the instructions and reliably use the response scale, then assess trait stability.

The model documentation provides context for this pattern. Llama 3.2 lists eight officially supported languages, excluding Burmese [6]. SEA-LION explicitly includes Burmese and English among 13 supported languages and was continued-pretrained and instruction-tuned for Southeast Asian use [7]. The present experiment does not identify which training component caused the difference, but the results are consistent with regional language support being important for structured Burmese tasks.

SEA-LION's complete profiles suggest broad cross-language stability under this prompt. Four trait means differed by no more than 0.08 points; Agreeableness differed by 0.43 points. The overall mean absolute difference of 0.14 is small relative to the four-point span of the response scale. Nevertheless, this descriptive similarity should not be interpreted as evidence of an internal or human-like personality. It shows that the model generated similar questionnaire response patterns across the two prompt languages under one experimental configuration.

The Qwen Burmese profile illustrates the danger of aggregation without completeness checks. Only 7.9% of item outputs were parseable. Depending on which items happened to yield a digit, session-trait means could be based on very few observations. This is reflected in standard deviations

as high as approximately 1.9 points. A chart can connect these means into a profile, but the visual shape does not overcome the missing-data problem. Llama's 100% failure rate correctly prevents any Burmese profile from being calculated.

4.2 Implications

For multilingual model evaluation, output validity should be treated as a primary metric rather than a preprocessing detail. A model may appear competent in open-ended text generation yet fail a constrained response task in a lower-resource language. Reporting only valid-score averages hides this failure and can reverse conclusions about cross-lingual consistency.

For practitioners selecting compact local models, parameter count alone is insufficient. The 8B SEA-LION model is larger than the two 3B models, so the study cannot separate size from regional specialization. However, the result supports including language-specific instruction-following tests in model selection. A small benchmark that reflects the intended language and output schema may be more informative than a broad English benchmark.

For psychometric evaluation of LLMs, the study reinforces cautions in prior work. Personality questionnaires can reveal stable patterns under some prompting conditions [3], and recent studies continue to compare questionnaire-derived profiles across models [4]. Yet validity depends on administration, prompt format, model capability, and scoring decisions. Generated scores should be described as behavioral properties of outputs, not direct measurements of a model mind.

4.3 Limitations

Several limitations constrain the findings. First, the English and Burmese prompts were parallel in purpose but cannot be assumed psychometrically equivalent without independent translation validation, back-translation, and review by multiple bilingual experts. Translation choices may alter tone, intensity, or cultural interpretation.

Second, the parser accepted the first recognizable scale digit anywhere in an output. This preserved more responses than strict exact matching but may count partially noncompliant text as valid. Future work should report both strict and permissive compliance rates and classify common failure modes.

Third, the models differed in size, model family, instruction tuning, language coverage, and possibly quantization. The design cannot attribute SEA-LION's performance to one factor. A controlled comparison would include multiple parameter sizes, quantization levels, and regionally adapted variants from the same base architecture.

Fourth, temperature 0.7 and 20 sessions capture only one region of sampling behavior. Results may change with deterministic decoding, system prompts, chat templates, reordered items, longer generation limits, or alternative scale wording. Although items were evaluated statelessly to prevent conversational carryover, stateless administration also differs from a human completing a questionnaire as one coherent session.

Fifth, severe Burmese missingness prevents fair trait comparison for two models. No imputation

was attempted because missingness was clearly model- and language-dependent rather than random. Finally, IPIP-50 measures broad domains only. Conclusions may differ with another questionnaire, narrower facets, behavioral tasks, or human evaluation of generated text.

4.4 Recommended future work

Future experiments should validate the Burmese translation with bilingual reviewers and document translation decisions. A short response-format calibration should run before the full 50-item evaluation. Models that fail could receive a culturally and linguistically appropriate few-shot example, while the original zero-shot condition remains as a baseline.

The evaluation should also store model digests, Ollama versions, hardware information, and random seeds. Strict exact-match parsing, permissive digit parsing, and semantic recovery should be reported separately. Confidence intervals or hierarchical models could then be applied only to conditions with adequate valid data. Finally, a broader model panel would help distinguish parameter scale from Southeast Asian language specialization.

5. Conclusion

This two-month internship project developed and applied a complete pipeline for cross-lingual IPIP-50 evaluation of three compact open language models. The system collected 6,000 item responses, recovered safely from interrupted runs, reverse-scored questionnaire items, summarized repeated sessions, audited parse failures, and generated publication-ready visualizations.

The central finding is that multilingual persona comparison depends first on multilingual instruction following. English responses were fully parseable for all models, but Burmese responses were nearly absent for Qwen 2.5 3B and entirely absent for Llama 3.2 3B. SEA-LION v3.5 8B-R produced complete data in both languages and showed a mean absolute trait difference of 0.14 points, with Agreeableness accounting for the largest language shift.

These results do not establish that an LLM possesses a human personality. They show that questionnaire response patterns and format compliance can vary sharply by language and model. The practical recommendation is clear: audit response validity, preserve raw outputs, and disclose missingness before interpreting cross-lingual psychometric profiles.

References

- [1] Goldberg, L. R. (1992). The development of markers for the Big-Five factor structure. *Psychological Assessment*, 4(1), 26-42. <https://doi.org/10.1037/1040-3590.4.1.26>
- [2] Oregon Research Institute. (n.d.). Administering IPIP measures, with a 50-item sample questionnaire. International Personality Item Pool. Accessed August 26, 2026. https://ipip.ori.org/new_ipip-50-item-scale.htm
- [3] Serapio-Garcia, G., Safdari, M., Crepy, C., Sun, L., Fitz, S., Romero, P., Abdulhai, M., Faust, A., & Mataric, M. (2023). Personality traits in large language models. arXiv:2307.00184. <https://arxiv.org/abs/2307.00184>
- [4] Bhandari, P., Naseem, U., Datta, A., Fay, N., & Nasim, M. (2025). Evaluating personality traits in large language models: Insights from psychological questionnaires. arXiv:2502.05248. <https://arxiv.org/abs/2502.05248>
- [5] Qwen Team. (2024). Qwen2.5 technical report. arXiv:2412.15115. <https://arxiv.org/abs/2412.15115>
- [6] Meta AI. (2024). Llama 3.2 3B Instruct model card. Hugging Face. <https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct>
- [7] AI Singapore. (2025). Llama-SEA-LION-v3.5-8B-R model card. Hugging Face. <https://huggingface.co/aisingapore/Llama-SEA-LION-v3.5-8B-R>

Appendix A. Reproducibility and Internship Deliverables

The repository contains the complete item-level dataset, derived session scores, summary statistics, questionnaire files, pipeline source, analysis source, and figures used in this paper. The main workflow is:

1. Install the Python dependencies listed in requirements.txt and start Ollama.
2. Download the three configured model identifiers.
3. Run `llm_pipeline.py` to collect or resume item-level responses.
4. Run `analyze_results.py` to calculate session traits and summary statistics.
5. Run `visualize_persona_results.py` to regenerate all figures.

Artifact	Purpose
<code>ipip-50_english.json</code>	English questionnaire items and scoring keys
<code>ipip-50_burmese.json</code>	Parallel Burmese items and scoring keys
<code>llm_pipeline.py</code>	Resumable stateless inference pipeline
<code>llm_persona_results.csv</code>	All 6,000 raw item-level model outputs
<code>analyze_results.py</code>	Reverse scoring and statistical summaries
<code>llm_persona_trait_scores.csv</code>	Per-session trait means
<code>llm_persona_summary.csv</code>	Model-language-trait means and standard deviations
<code>visualize_persona_results.py</code>	Publication figure generation

The internship deliverables demonstrate experimental design, multilingual dataset preparation, local model inference, fault-tolerant data collection, psychometric scoring, missing-data auditing, statistical reporting, visualization, technical documentation, and reproducible research packaging.